

Bringing Orbital TIG Welding into an Industrial Piping Workshop

Welding a pipe that cannot be turned

Nathan JUST | Apprentice Engineer, Mechanical and Industrial Structures (MSI) | Class of 2023–2026

Academic tutor: **Éric ANGLARET** | Company tutor: **Hugo PAGANO**

1 Where I worked



PHOTO

A typical piece of work by the company
(to be supplied)

CIMAT SARTEC

FOSELEV GROUP

Industrial piping and maintenance for the chemical, petrochemical and steel industries. I worked there as an apprentice project engineer, following jobs from the first estimate through to installation on site. That is how this study landed in my hands.



Piping



Boilermaking

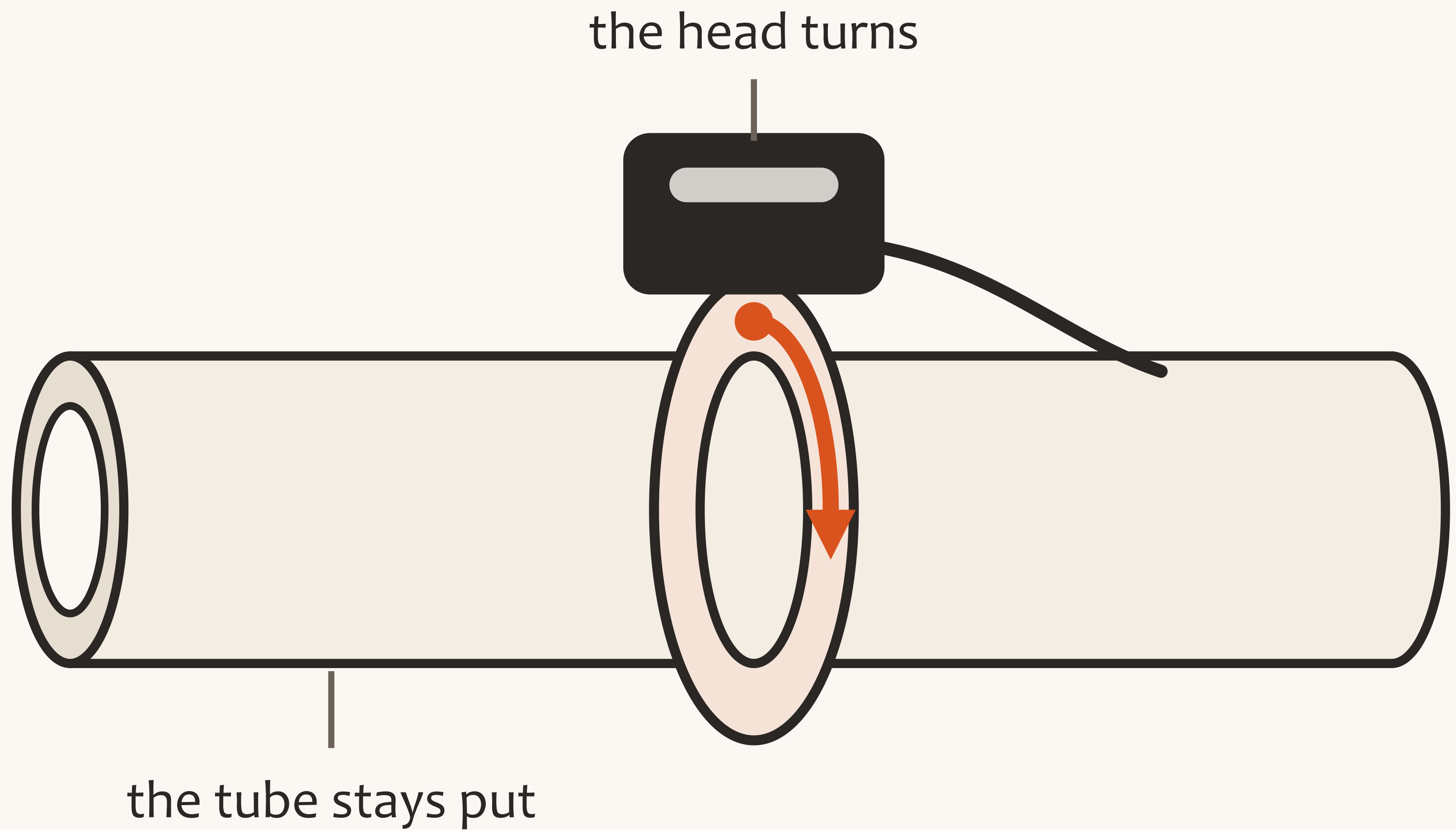


Steel structures

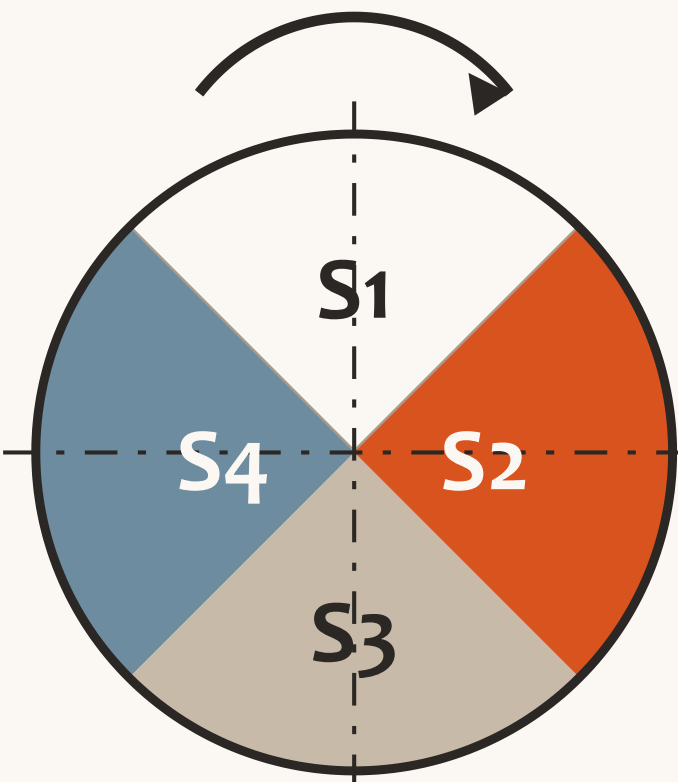
Les Angles (30), France. Part of a group of around 3,300 people, started in 1970.

2 How it works

Every pipe weld in the workshop is still made by hand, and a manual weld depends on the person holding the torch. Meanwhile customers keep asking for cleaner welds, welds that come out the same every time, and proof of how each one was made.

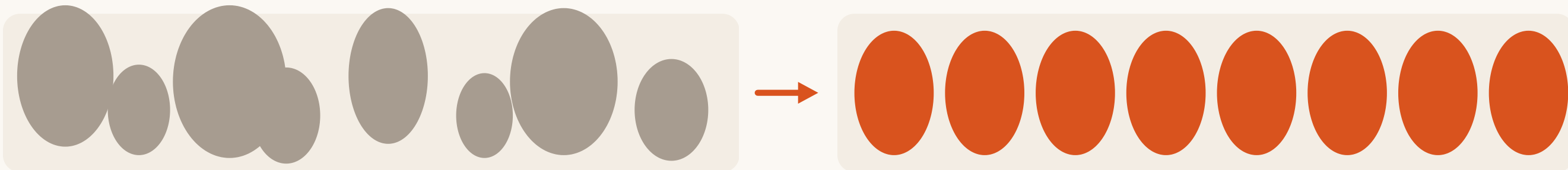


The head clamps onto the tube and carries the torch right around the joint, on its own. The welder sets the programme, then watches it run.

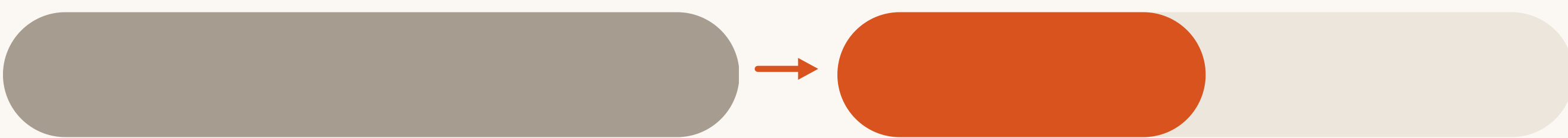


One turn, four sectors, four sets of settings: flat, vertical down, overhead, vertical up. Sector by sector, the molten pool never gets the chance to run.

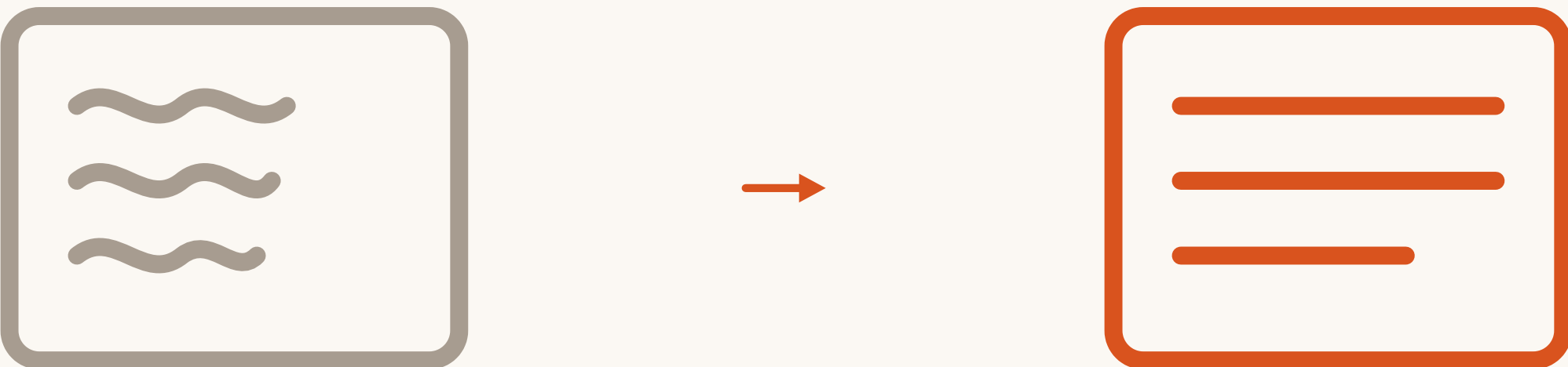
3 What would change



A bead that comes out the same every time
The pulsed current holds the pool, whatever the position.



Roughly half the welding time
On the joint taken as a reference, times still to be measured.



Every weld recorded by the machine, not by hand
The traceability customers now ask for comes with the process.

Nothing in the standards stands in the way. The welding procedure is qualified first, then the operators to ISO 14732, then the usual inspections. The road is well marked out.

RECOMMENDATION

Worth the investment.

Welding times to be confirmed in the workshop first.

Next: out on site with the teams, and an open head fed with wire for the bigger pipes. And the programmes built up job after job slowly become a know-how of their own, one that gains value instead of wearing out.